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Order flow analysis of cryptocurrency markets

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rates in cryptocurrency markets are found to be the main differentiators between the nascent asset class market microstructure and that of the established markets.

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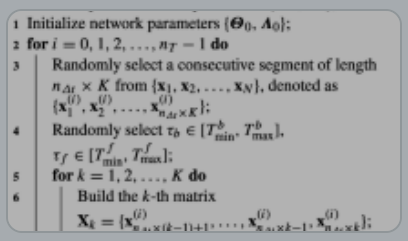

Initialize network parameters $\{\Theta_0, \Lambda_0\}$;
for $i = 0, 1, 2, \ldots, n_T - 1$ **do**
  Randomly select a consecutive segment of length $n_{\Delta t} \times K$ from $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$, denoted as $\{\mathbf{x}_1^{(i)}, \mathbf{x}_2^{(i)}, \ldots, \mathbf{x}_{n_{\Delta t} \times K}^{(i)}\}$;
  Randomly select $\tau_b \in [T_{\min}^b, T_{\max}^b]$, $\tau_f \in [T_{\min}^f, T_{\max}^f]$;
  for $k = 1, 2, \ldots, K$ **do**
    Build the k-th matrix $\mathbf{X}_k = \{\mathbf{x}_{n_{\Delta t} \times (k-1)+1}^{(i)}, \ldots, \mathbf{x}_{n_{\Delta t} \times k-1}^{(i)}, \mathbf{x}_{n_{\Delta t} \times k}^{(i)}\}$;

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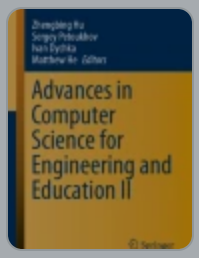

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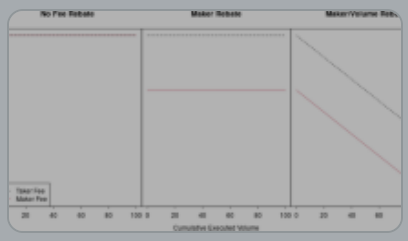

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feature of BitMex is leverage that it offers to traders. Currently, maximum leverage that one can take out on XBTUSD contract is \(\times \) 100.

XBTUSD is effectively a perpetual swap contract, where one contract is worth 1 USD of Bitcoin. XBTUSD never expires, but participants are may be subject to margin funding. The contract tracks the underlying price of Bitcoin, which is calculated as an index across various spot markets. The tracking mechanism is dependent upon funding ratio. In essence, to reduce tracking error, BitMex will calculate the deviation between current XBTUSD contract value and spot price index. If the value of the contract is above the reference index, than the implied interest rate of Bitcoin is higher that USD. Hence, to stabilise the price, the long contract holders will pay funding the short-sellers of the contract. This mechanism applies vice versa when contract value falls below the reference index and is what keeps the contract at fair price.

Trading fee structure on BitMex is very straightforward and highly shifted towards market makers when compared to other exchanges. Market makers get paid a constant 25 bps rebate, while takers pay 35 bps in commission.

Appendix B: Python code

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Listing 1: Python Code for Data Import, Model Fitting and Stationarity Tests

```
import statsmodels.api as sm
from statsmodels.regression.linear_model import OLS
import statsmodels.tsa.stattools as ts
import pandas as pd
import numpy as np
%matplotlib inline
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = (12.0, 10.0)

def load_data_hdf(path, date_from, date_to = None):
    df = pd.read_hdf(path, index_col = 'timestamp')
    df.index = pd.to_datetime(df.index.values, format='%Y-%m-%dD%H:%M:%S.%f000')
    return df.loc[date_from:]

def ofi(quotes, dt):
    qdf = reset_index(quotes.copy())

    qdf['Mid-Price_Change'] = ((qdf['bidPrice'] + qdf['askPrice'])/2).diff().div(0.1)

    qdf['prev_bidprice'] = qdf['bidPrice'].shift()
    qdf['prev_bidsize'] = qdf['bidSize'].shift()
    qdf['prev_askprice'] = qdf['askPrice'].shift()
    qdf['prev_asksize'] = qdf['askSize'].shift()

    qdf.dropna(inplace=True)

    bid_geq = qdf['bidPrice'] >= qdf['prev_bidprice']
    bid_leq = qdf['bidPrice'] <= qdf['prev_bidprice']
    ask_geq = qdf['askPrice'] >= qdf['prev_askprice']
    ask_leq = qdf['askPrice'] <= qdf['prev_askprice']

    qdf['OFI'] = pd.Series(np.zeros(len(qdf)))
    qdf['OFI'].loc[bid_geq] += qdf['bidSize'].loc[bid_geq]
```


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```
def set_nanosecond_index(df):
    tmp = df.copy()
    tmp.set_index('index', inplace=True)
    tmp.index = pd.to_datetime(tmp.index.values, format='%Y-%m-%dD%H:%M:%S.%f000')
    return tmp

def reset_index(df):
    tmp = df.copy()
    return tmp.reset_index()

def plot_ofi_and_ols_summary(df, dt):
    df.plot(kind='scatter', grid=True,
            x='OFI', y='Mid-Price Change',
            title = 'XBTUSD '+ dt + ' OFI',
            alpha=0.5, figsize=(12,10))
    ofi_ = sm.add_constant(df['OFI'])
    ols = OLS(df['Mid-Price Change'], ofi_).fit()
    print(ols.summary2())



def plot_tfi_and_ols_summary(tf, dmid, dt):
    pd.concat([tf,dmid], axis = 1).plot(kind='scatter', grid=True,
                                        x='TFI', y='Mid-Price Change',
                                        title = 'XBTUSD '+ dt + ' TFI',
                                        alpha=0.5, figsize=(12,10))
    tf_ = sm.add_constant(tf)
    ols = OLS(dmid, tf_).fit()
    print(ols.summary2())

def adf_test(var):
    adf = ts.adfuller(var, 1)
    print('ADF Statistic: %f' % adf[0])
```


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